

- What is your understanding of Code collaboration? How do you collaborate with your group in coding? What are the common tools/platform to support the Code collaboration?

Part I Data Acquisition and Manipulation

There are **8** questions in this part, totalling **60** marks. Each of question is worth **5** marks. Additionally, the quality of your explanation in both the report and video will collectively be worth **20** marks.

You are recommended to use Google Colab to finish all the coding in the *code block cell*, and **provide sufficient coding comments**, and also **save the result of running as well**.

The (business_review_submission.zip) data used for this part could be found in **here**. There are two files in the data. The first one is about the business review submission with many companies. For each of the row, the review submission is provided with relevant information such as user id, time, name and many others. The second one is the meta information of the business and the two data could be joined with gmap_id. You will need to use spark to **first read the unzipped (csv) review data for starting** and later join the meta review business data on dataframe (pandas or spark). You could find the code on reading csv data with Spark from M04G. In some of the tasks, if the question is not specifically asking to use spark, you could use both pandas and numpy.

Question 1.1

Using PySpark to do some data wrangling process, so that:

1.1.1 For the none or null in text column, change it to 'no review'.

1.1.2 Process the content in time column, and convert the strings from time to yyyy-mm-dd format in the new column as newtime and show the first 5 rows.

Question 1.2

Find out the information for gmap_id on the reviews. In order to achieve the above, some wrangling work is required to be done:

1.2.1 Using pyspark to calculate the number of reviews per each unique gmap_id and save as float format in pyspark dataframe to show the top 5 rows.

1.2.2 Transform the current pyspark dataframe to pandas dataframe (named as df) and create the column reivew_time with the information of review time on hours level. Print your df pandas dataframe with top 5 rows after creating the column review_time.

1.2.3 Using matplotlib or seaborn to draw some (two or more if possible) visualizations on the relationship between gmap_id and reivew_time. You could explore for example, what is the time people usually review? How many business is reviewed in the morning time etc. Please also discuss the insights you are finding with your visualizations in the markdown cell. Please also include your findings and visualizations in the report.

Question 1.3

Let's continue to analyze the reivew_time with reviews and related gmap_id. You need to use another data meta-business to join with the current dataframe on gmap_id.

1.3.1 Determine which workday (day of the week), generates the most reviews (plotting the results in a line chart with workday on averaged submissions).

- 1.3.2** Identify the names of business (column name from data meta-business) that has the highest averaged ratings on 'that workday' (you need to find out from 1.3.1), and find out which category those businesses are from?
- 1.3.3** Please further explore the data on name of business and find out some more insights by yourself such as which category it is and what are the peak hours etc. Please use visualizations and tables to support your findings and write down the insights in the markdown cell. Please also include your findings and visualizations in the report.

Question 1.4

For the reviews on each of the submissions, work on all the review content and find out the top 30 most common words; Also generate separate word cloud visualizations for different years by grouping the reviews by review year and write down the insights in the markdown cell. Please also include your findings and visualizations in the report.

Question 1.5

Let's do some analysis on the business_name and the reviewers. Determine the number of unique reviewers of business and its categories to identify which business / category has attracted the most reviewers (find out the highest distinct count of reviewers on business / category level). Also, analyze the temporal patterns of when reviewers submitted their reviews (you could leverage the workday, year, month, or hours to conduct the analysis) and share your findings and insights in the markdown cell. Please also include your findings and insights (visualizations) in the report.

Question 1.6

As the data scientist, you are required to build a recommendation for the business by using reviews, ratings, and its categories. In this task, you need to:

- 1.6.1** Write down your strategy of building the recommendation on business for customers in the markdown cell. You could create your own strategy or leverage the provided one here KNN on collaborative filtering. Please also include your strategy details in the report.
- 1.6.2** Could you please try to implement the strategy (code) you have written down for the recommendation system? Please give detailed explanation of your code and the logic in the comments and also interpret the recommendations with examples in the markdown cell. Please also include your implementation details and results in the report.

Question 1.7

Continue work on the review data you have now, for each of the submissions of the review, you will need to explore the rating with other information:

- 1.7.1** Build visualization to explore the relationships of the rating and business categories. Please write down your insights in the markdown cell and also include your insights and visualizations in the report.
- 1.7.2** Let's focus on the lower ratings now. Could you please find out the actual reviews on lower ratings and analyze on the reason? (You could use the common used words in lower rating reviews or design your own strategy with reasonable logic). Please also include your analysis details in the report.

Question 1.8

Continue to work on the submission of the reviews, we would like to focus on the reviewer level with all the reviewed business history, but before we actually conduct the programming, we will need to finish few questions for exploration:

- 1.8.1** Check on the reviewer level reviewed business, sort the review of each business by the review time (newtime column) and then save the business name into the list variable `user_business_list` for each reviewer.
- 1.8.2** Check on the `user_business_list`, could you observe some repeated business names for the same user? If so, could you remove those duplicated business names under same user? Please print out the number of element in the `user_business_list` for each reviewer before removing the duplicated business name and after removing the duplicated business name.
- 1.8.3** Check on the `user_business_list`, could you find the user similarities according to their past reviewed business ? You are free to design your own strategy and give sufficient explanation in markdown cell and code implementation together. Please also include your strategy details and implementation in the report.

Hint: you might consider to use encoding for each of the business names and then calculate the difference of the users.